

MAINUDDIN KHUDAVAND

AI/ML Engineer | Full Stack Developer (MERN)

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PROFILE SUMMARY

AI/ML Engineer and Full Stack Developer focused on Deep Learning and Agentic AI. Experienced in designing and building intelligent, AI-driven systems — from LLM-powered agents to end-to-end ML pipelines — paired with scalable MERN stack applications. Delivered multiple projects at the intersection of machine learning and full-stack engineering, including a filed patent for a role-based digital governance platform. Passionate about turning applied ML and agentic-systems research into usable products.

EDUCATION

B.E in Computer Science and Engineering

Sep 2023 – July 2027

Tontadarya College of Engineering, Gadag | CGPA: 8.5

PROJECTS

Empowering Villages Through Digital Governance (MERN)

Live Demo: <https://empowering-villages-through-digital-6r80.onrender.com>

- Developed a full-stack, five-role platform — citizens, field workers, department officials, district managers, and admins — enabling complaint filing, task tracking, and workflow oversight via role-specific dashboards.
- Integrated secure authentication, role-based access control, and SMTP-based notification logic across all five roles to support seamless, escalation-aware multi-role interaction.

Tech Stack: Node.js, Express.js, React.js, MongoDB, JWT Authentication, SMTP, REST APIs

Intelligent AI Attendance — Face & Voice Recognition

Live Demo: <https://ai-attendance-project-landing-main1-1.onrender.com>

- Built an AI-powered attendance system combining facial recognition and voice authentication for secure, contactless, dual-factor identity verification.
- Designed separate teacher and student portals with QR code enrollment, subject management, and automated attendance logging directly from classroom photos and audio recordings.
- Trained an SVM classifier on face embeddings and used Resemblyzer voice embeddings for speaker verification, with bcrypt-hashed credentials and a Supabase (PostgreSQL) backend for persistent, secure storage.

Tech Stack: Html, Css, Js, Flask, Python, Streamlit, Supabase (PostgreSQL), dlib, face & voice recognition_models, Scikit-learn (SVM), Resemblyzer, librosa, segno

Real-Time AI Gym Trainer

Live Demo: <https://ai-gym-trainer-landing-page.onrender.com>

- Built an AI-powered fitness assistant that uses computer vision to detect human posture and provide real-time corrective feedback during workouts.
- Integrated deep learning models for exercise recognition, rep counting, and form validation, enabling personalized training guidance without human supervision.

Tech Stack: Python, Streamlit, WebRTC, MediaPipe, OpenCV, Groq LLM, gTTS, SQLite

TECHNICAL SKILLS

Languages & Databases: JavaScript, Python, SQL, MongoDB

Web Development: Node.js, Express.js, React.js, REST APIs, Flask, Streamlit, Supabase

Machine Learning / Deep Learning: Scikit-learn, PyTorch, Pandas, NumPy, Regression, Classification, FNN, CNN, RNN, Transformers, GAN, DQN, SARSA

Generative AI: LLMs, RAG Pipelines, Agentic AI, OpenAI API, NLP

Data Visualization: Seaborn, Matplotlib

DevOps & Tools: Git & GitHub, Docker, CI/CD, Kubernetes, Render, Vercel, Jest (Unit Testing)

Computer Science Fundamentals: Computer Networks, Operating Systems, DBMS, OOPs

CERTIFICATIONS & ACHIEVEMENTS

- Selected for Navitantriks 2026, a state-level competition, for the project "Empowering Village through Digital Governance."
- Patent (Filed): "A System and Method for Digital Governance and Grievance Redressal Management in Rural Administrative Units Using a Role-Based Web Platform" — Application No. 202641040564, filed 31/03/2026. Status: Published (Awaiting Examination).
- Empowering Residency — registered under Udyam (MSME), Govt. of India. Udyam Reg. No. UDYAM-KR-14-0034950, Micro Enterprise, Services Sector.
- Apna College Certificate — Full Stack Web Development (2025)
- Apna College Certificate — AI/ML (2026)
- State Level Competition Certificate